Vecteurs Gaussiens

- $Var(aX+bY)$, $a, b \in \mathbb{R}$

- $Var(aX+bY) = a^2 var(X) + b^2 var(Y) + 2ab\ Cov(X,Y)$

avec $cov(X,Y) = E(X.Y) - E(X).E(Y)$

- $Cov(aX, bY) = a.b\ cov(X.Y)$
- $Cov(X,X) = Var(X)$
- $Cov(\alpha_1 X + \alpha_2 Y, \beta_1 X + \beta_2 Y) = Cov(\alpha_1 X, \beta_1 X + \beta_2 Y) + Cov(\alpha_2 Y, \beta_1 X + \beta_2 Y)$

$= Cov(\alpha_1 X, \beta_1 X) + Cov(\alpha_1 X, \beta_2 Y) + Cov(\alpha_2 Y, \beta_1 X) + Cov(\alpha_2 Y, \beta_2 Y)$

$= \alpha_1 \beta_1 Var(X) + \alpha_2 \beta_2\ cov(X,Y) + \alpha_2 \beta_1\ Cov(Y,X) + \alpha_2 \beta_2\ Var(Y)$

$= \alpha_1 \beta_1 Var(X) + (\alpha_1 \beta_2 + \alpha_2 \beta_1)\ Cov(X,Y) + \alpha_2 \beta_2\ Var(Y)$